Marlon Munoz

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Education

2018-2022 **Physics/Astro B.S, Santa Barbara, *University of California: Santa Barbara***

Research Experience

Apr 2020 - Jun 2020

Independent Research - Deborah Fygenson Student Lab

- Observed the correlation between candle burning and indoor pollution.
- Utilized programming, sensors, and Raspberry Pi to take measurements.

Jan 2022 - May 2022

Remote Lab Assistant - Deborah Fygenson Remote Labs

- Improved remote labs by implementing multi-processing in order to allow equipment to operate more efficiently.
- Provided tech support and helped repair hardware in remote labs.

Jan 2022- Aug 2022

Research Assistant - Galan Moody Lab

- Helped develop a boson sampling simulation in order to simulate four photon graph states .
- Designed a deep learning algorithm that models and significantly reduces the noise sources of the four photon graph states.

Apr 2021 - Current

Research Assistant - Philip Lubin Lab

- Implements deep learning algorithms in order to improve detection of transients in sky survey
- Programs microcontrollers for sensor readout and laser communication.
- Created simulations of meteor impacts using python and mathematica.
- Utilized machine learning algorithms that recognize objects and provide positional vector quantities in real time.

Relevant Coursework

- **Statistics, Data Analysis, and Machine Learning for Physicists**
- **Quantum Mechanics**
- **Data Science Principles/Statistics**
- **Problem Solving With Computers (C++)**
- **Thermal and Statistical Physics**
- **Optics and Electromagnetism**

Skills

Python • C++ • MatLab • LaTeX • Machine Learning
Multi-Processing • Raspberry Pi • Mathematica • Data Science